

QUETTA INTL, Pakistan

WMO: 416600

Lat: 30.251N

Lon: 66.938E

Elev: 5250

StdP: 12.11

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.7	25.4	7.5	9.8	37.1	11.2	11.8	36.4	28.2	39.7	25.4	44.7	3.6	130	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	99.6	67.6	97.9	67.0	96.1	66.2	72.3	93.4	70.6	91.9	69.1	90.5	16.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	116.2	85.7	63.7	107.7	84.3	61.6	99.7	84.5	39.9	95.0	38.1	92.5	36.7	90.2	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.3	24.4	22.0	DB	17.7	102.6	4.5	0.8	14.5	103.1	11.9	103.6	9.4	104.0	6.1	104.6	(4)
(5)			WB	15.2	75.5	3.9	4.1	12.3	78.4	10.0	80.8	7.8	83.1	4.9	86.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	41.4	45.7	56.0	65.6	74.5	81.8	85.6	82.1	74.0	63.4	51.5	43.4		
	DBStd	16.26	6.23	7.48	6.69	6.00	4.82	3.96	2.57	2.98	4.76	5.80	6.11	5.96		
	HDD50	722	272	159	22	0	0	0	0	0	0	0	54	215		
	HDD65	2801	730	539	284	64	3	0	0	0	2	104	406	669		
	CDD50	5784	7	40	210	470	759	955	1102	997	721	415	98	10		
	CDD65	2388	0	0	7	83	297	505	637	532	274	53	0	0		
	CDH74	30777	0	3	114	1037	3704	6785	8685	6517	3135	784	12	1		
	CDH80	15863	0	0	12	301	1724	3777	4984	3493	1369	203	0	0		
(14) Wind	WSAvg	10.3	8.3	10.2	10.8	11.6	12.4	12.8	13.8	11.8	9.2	8.1	7.5	7.3		
(15) Precipitation	PrecAvg	9.1	1.8	1.7	1.9	1.2	0.3	0.1	0.4	0.2	0.1	0.2	0.3	0.9		
	PrecMax	16.5	7.0	7.4	9.1	5.8	1.6	1.7	4.8	1.1	3.7	2.7	2.6	5.7		
	PrecMin	2.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.3	1.5	1.5	1.7	1.4	0.4	0.2	0.9	0.3	0.6	0.5	0.5	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.2	72.8	81.5	89.6	96.7	100.9	102.1	100.1	94.2	88.1	75.7	67.7		
		MCWB	47.4	50.4	59.1	60.9	63.7	66.2	69.6	68.7	64.0	59.7	54.3	48.1		
	2%	DB	61.5	67.8	77.0	86.4	94.1	98.9	100.2	97.5	91.6	85.1	72.3	63.3		
		MCWB	46.0	49.2	57.0	60.2	62.9	65.9	68.5	67.5	62.4	58.5	52.9	46.2		
	5%	DB	57.9	63.6	73.8	83.7	91.8	97.0	98.6	95.4	89.7	82.2	69.2	60.0		
		MCWB	43.9	47.6	55.1	59.4	62.6	65.3	67.7	66.4	61.7	57.3	51.2	44.3		
	10%	DB	54.2	59.3	70.0	80.2	89.2	94.8	96.8	93.5	87.7	78.6	66.0	56.6		
		MCWB	42.0	45.2	53.1	57.6	61.8	64.3	67.4	65.3	61.5	55.5	49.8	43.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	53.0	63.3	66.3	70.8	72.5	75.5	73.2	70.1	64.1	57.9	51.4		
		MCDB	61.3	67.5	76.5	82.0	90.4	94.1	97.5	94.4	88.2	83.5	71.9	63.7		
	2%	WB	47.0	50.7	59.1	63.3	67.5	70.3	73.1	71.0	67.3	60.6	54.8	47.8		
		MCDB	58.4	64.8	74.1	81.4	88.5	91.3	94.7	91.9	85.9	80.4	67.6	59.6		
	5%	WB	45.1	48.6	56.2	61.2	65.6	68.7	71.1	69.4	65.6	58.5	52.6	45.7		
		MCDB	55.5	61.0	71.3	80.1	87.4	90.5	93.0	90.5	83.8	79.0	66.1	57.0		
	10%	WB	43.2	46.7	53.9	59.2	63.6	67.1	69.5	67.8	63.7	56.7	50.7	43.6		
		MCDB	52.6	58.3	68.4	77.6	85.3	89.8	91.8	89.0	81.8	76.2	64.1	54.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.8	20.7	22.0	24.7	26.1	25.0	23.0	23.5	27.1	28.2	25.7	23.5		
		MCDBR	27.4	25.6	27.4	28.6	30.0	27.0	25.4	25.3	27.8	30.0	28.8	27.9		
		MCWBR	15.8	12.8	14.2	12.7	12.0	9.1	8.4	8.8	10.2	11.8	14.5	15.3		
		MCDBR	23.5	23.4	26.0	27.0	28.8	24.6	24.1	24.2	25.9	29.7	26.2	24.2		
	5% WB	MCWBR	14.6	12.8	15.0	14.5	14.2	10.2	10.0	10.3	11.8	14.4	14.4	15.1		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.401	0.449	0.507	0.505	0.531	0.645	0.582	0.431	0.371	0.349	0.313		
	taud		2.238	2.031	1.894	1.792	1.798	1.744	1.557	1.667	1.971	2.108	2.184	2.322		
	Ebn at Noon		281	269	264	253	254	246	219	231	266	275	272	284		
	Edh at Noon		36	49	61	69	69	73	88	78	55	45	38	32		
(44) All-Sky Solar Radiation	RadAvg		1183	1387	1832	2165	2452	2574	2357	2239	2115	1807	1391	1181		
	RadStd		96	144	125	113	98	100	120	118	69	46	65	75		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page